

**Practice Final**

**Name:** \_\_\_\_\_

**ID:** \_\_\_\_\_

**Score:** \_\_\_\_\_

| Problem | Pts. | Score |
|---------|------|-------|
| 1       | 30   |       |
| 2       | 30   |       |
| 3       | 40   |       |
| 4       | 30   |       |
| Total   | 130  |       |

**I certify that this is my sole work and I haven't discussed any problem with other person.**

**Signature:** \_\_\_\_\_

1. Please print your name and student ID number.
2. Sign on the above statement.
3. For full credit, you should show all of your work and explain your reasoning.

**GOOD LUCK!**

**Problem 1 (30 Pts)** Let  $g[n] = (-1)^n h[n]$ , where  $h[n] = \frac{\sin(\pi n/2)}{\pi n}$  whose DTFT satisfies

$$H(e^{j\omega}) = \begin{cases} 1 & \text{if } |\omega + 2\pi k| < \frac{\pi}{2} \text{ for } k \in \mathbb{Z} \\ 0 & \text{else.} \end{cases}$$

- (a) (10 Pts) Draw  $|G(e^{j\omega})|$  where  $G(e^{j\omega})$  denotes the DTFT of  $g[n]$ . Explain how you obtained the graph.

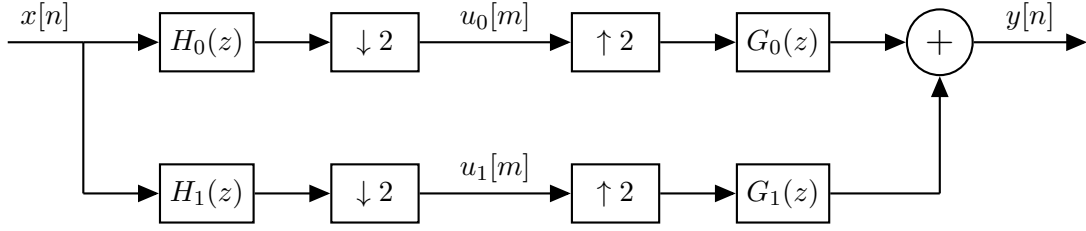
- (b) (10 Pts) Let  $f[n]$  be obtained by up-sampling  $g[n]$  by factor 2, that is,

$$f[n] = \begin{cases} g[n/2] & \text{if } n/2 \in \mathbb{Z} \\ 0 & \text{else.} \end{cases}$$

Draw  $|F(e^{j\omega})|$  where  $F(e^{j\omega})$  denotes the DTFT of  $f[n]$ . Explain how you obtained the graph.

- (c) (10 Pts) Let  $v[n]$  be obtained by  $V(z) = H(z)F(z)$ , where  $V(z)$  and  $H(z)$  respectively denote the  $z$ -transform of  $v[n]$  and  $h[n]$ . Draw  $|V(e^{j\omega})|$  where  $V(e^{j\omega})$  denotes the DTFT of  $v[n]$ . Explain how you obtained the graph.

**Problem 2 (40 Pts)** We consider the following two-channel filterbank system.



- (a) (10 Pts) Let  $g_0[n] = \frac{1}{\sqrt{2}}\delta[n] + \frac{1}{\sqrt{2}}\delta[n-1]$  denote the impulse response of  $G_0(z)$ . Determine whether  $g_0[n]$  has generalized linear phase or not. Provide justification for full credit.

- (b) (10 Pts) Let the other three filters be obtained from  $G_0(z)$  by the following identities:

$$G_0(z) = H_1(-z), \quad G_1(z) = -H_0(-z), \quad H_1(z) = -z^{-1}H_0(-z^{-1}).$$

Compute the impulse responses  $g_1[n]$ ,  $h_0[n]$ , and  $h_1[n]$ . Provide derivation for full credit.

- (c) (10 Pts) Let the filters be obtained from the previous parts. Sketch the magnitude responses  $|G_0(e^{j\omega})|$ ,  $|G_1(e^{j\omega})|$ ,  $|H_0(e^{j\omega})|$ , and  $|H_1(e^{j\omega})|$ . Provide justification for full credit.

- (d) (10 Pts) Show that the entire system provides the perfect reconstruction up to delay, that is  $y[n] = x[n - l]$  for some non-negative integer  $l$ . (Hint: Compute  $Y(z)$  from  $U_0(z)$  and  $U_1(z)$ .)

**Problem 3 (40 Pts)** Let  $h[n]$  denote the impulse response of a discrete-time filter whose  $z$ -transform is given by

$$H(z) = (1 - z^{-1}) (1 - 2z^{-1}) \left(1 - \frac{1}{2}z^{-1}\right).$$

(a) (10 Pts) Draw the pole-zero diagram of  $H(z)$  and draw the impulse response of  $h[n]$ .

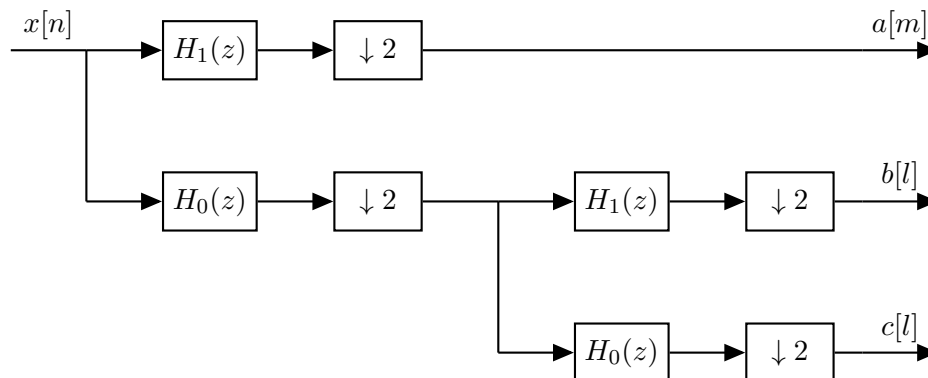
(b) (10 Pts) Determine whether  $h[n]$  is a generalized linear phase (GLP) filter by using the result of part (a).

(c) (10 Pts) Let  $G(z) = H(-z)$ . Determine whether  $G(z)$  corresponds to a GLP filter. Justify your answer.

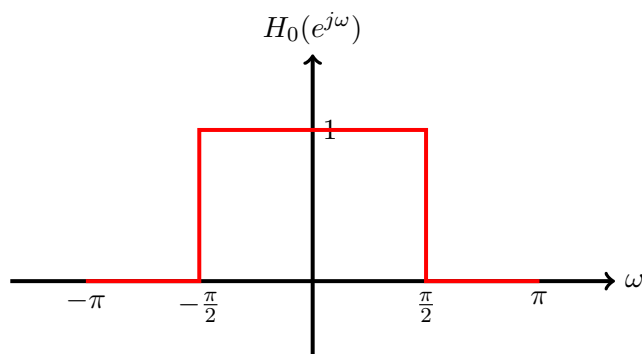
(d) (10 Pts) Let  $F(z) = G(z^{-1})$ . Determine whether  $F(z)$  corresponds to a GLP filter. Justify your answer.



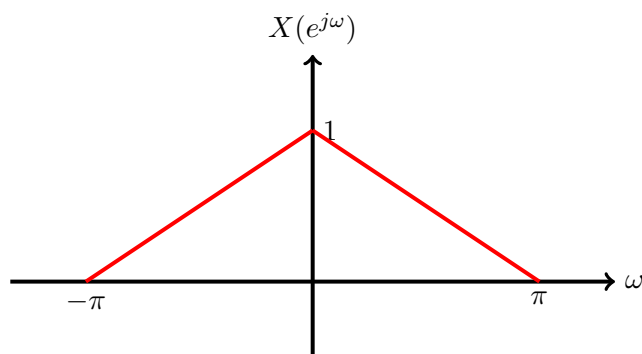
**Problem 4 (30 Pts)** We consider the following 2-level wavelet decomposition system.



Suppose that the DTFT of  $H_0[n]$  is given as in the following plot.



Suppose that  $H_1(e^{j\omega}) = H_0(e^{j(\omega-\pi)})$ . Draw the DTFT of output channels when the input DTFT is given as in the following graph.



Explain how you obtained the graphs.

